

## Air Quality Health Index (AQHI) — Edmonton

# 3.0

Current AQHI at Edmonton McCauley (0.84 km from venue)  
Forecast Sat 6:00 AM at Edmonton McCauley (0.84 km from venue): **3.4**

| Health Risk | AQHI | At-Risk Population                   | General Population                        |
|-------------|------|--------------------------------------|-------------------------------------------|
| Low Risk    | 3.4  | Enjoy your usual outdoor activities. | Ideal air quality for outdoor activities. |

No active Environment Canada weather alerts for the venue.

## Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 12.7°C and feels near 12.5°C. Winds are 3 km/h from NW, gusting near 6 km/h.
- The nearest current AQHI is 3 at Edmonton McCauley, 0.8 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 5 (forecast tonight: 2, tomorrow: 2).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 1.6 km away) puts the area near 1.
- The nearest air monitoring station (Edmonton McCauley, 0.8 km) reports fine particulate matter (PM2.5) at 42.5 µg/m³.
- A nearby community sensor ('RE/MAX Field', 1.2 km) reads 5.9 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 181.4 km to the W, i.e. arriving from the W.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-08-28) is 7.0 km E of the venue. The modeled back-trajectory passes within 13.7 km of this fire, making smoke transport to the venue plausible based on wind-field modeling.
- The AQHI forecast for Sat 6:00 AM is 3, based on Edmonton McCauley, 0.8 km from the venue.
- Overall operational environmental risk is LOW.

## Recommendations

- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

## Hazard Snapshot — Overall Risk: Low Risk

|                                                                       |                                                            |                                                                            |                                                             |                                                                   |
|-----------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------|
| <b>AIR QUALITY</b><br><b>Low Risk</b><br>3.4 AQHI                     | <b>HEAT</b><br><b>Low Risk</b><br>20.8 °C apparent/humidex | <b>WIND</b><br><b>Low Risk</b><br>22.3 km/h peak gust                      | <b>PRECIPITATION</b><br><b>Low Risk</b><br>0.0 mm/h maximum | <b>THUNDERSTORM</b><br><b>Low Risk</b><br>— first forecast signal |
| <b>AQHI RATE OF CHANGE</b><br><b>Low Risk</b><br>0.0 AQHI change/hour |                                                            | <b>WIND SHEAR</b><br><b>Low Risk</b><br>2.9 ° direction diff (10m vs 120m) |                                                             |                                                                   |

## Local Monitoring

|                                                              |                                                   |                                                                     |                                                                 |
|--------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|
| Nearest station<br><b>3.0</b><br>Edmonton McCauley · 0.84 km | Blended estimate<br><b>1.0</b><br>confidence High | PM2.5 (station)<br><b>42.5 µg/m³</b><br>Edmonton McCauley · 0.85 km | PM2.5 (community)<br><b>5.9 µg/m³</b><br>RE/MAX Field · 1.21 km |
|--------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------|

